MAHARSHI OZA

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PROFESSIONAL SUMMARY

Design Verification engineer with 4+ years of experience in digital IP verification and validation across ASIC/FPGA and automotive domains. Skilled in SystemVerilog, UVM, constrained-random verification, SVA, scoreboards, and functional/code coverage closure, with hands-on AXI4 and memory-subsystem verification. Strong foundation in RISC-V microarchitecture, out-of-order execution, and memory hierarchies.

CORE COMPETENCIES

Languages & Protocols: SystemVerilog, Verilog HDL, C++, Python, Perl, Shell Scripting, AXI4, I2C, RISC-V ISA
Design Verification: UVM, Constrained-Random Verification (CRV), SVA, Functional & Code Coverage, Coverage Closure, Regression Testing, Debug & Root-Cause Analysis, Scoreboards, TLM, SoC/IP Verification
RTL & ASIC: RTL Design, Logic Synthesis, Static Timing Analysis (STA), Clock Domain Crossing (CDC), Design for Testability (DFT)
Tools: VCS, Xcelium, QuestaSim, ModelSim, Design Compiler, Vivado, GDB, JTAG, Vector CANoe/CANalyzer, Git, Linux
Computer Architecture: RISC-V, Pipelining, Superscalar & Out-of-Order Execution, Register Renaming, Branch Prediction, Memory Hierarchy
GPU & AI Hardware: GPU/SIMT Architecture, CUDA, GPU Memory Hierarchy, Warp Execution, Parallel Computing, Matrix Multiplication, Convolution, Parallel Reduction, AI/LLM Accelerators

PROFESSIONAL EXPERIENCE

RTL Design Verification Engineer | Maxgen Technologies Pvt Ltd., India **Aug 2023 – Aug 2024**

- Implemented an out-of-order scoreboard for an AXI4 memory subsystem in SystemVerilog/UVM, tracking outstanding transactions by ID and checking read/write data, responses, and per-ID ordering rules.
- Developed functional coverage for AXI4 burst types, lengths, sizes, IDs, out-of-order completion, and handshake stalls, achieving over 90% functional coverage; maintained UVM environments for Asynchronous FIFO and FPU blocks.
- Verified corner cases including reset/recovery, FIFO full/empty and overflow/underflow, data integrity, and AXI handshake correctness; debugged regressions using Questa, Xcelium, VCS, and Vivado to isolate RTL and testbench issues.
- Contributed to the RTL design of a multi-port packet router (FSM control, internal FIFOs, synchronizers) and aligned verification scenarios with ongoing RTL and specification updates.

Functions Engineer – Vision & Perception | Magna Electronics, India **Dec 2021 – Aug 2023**

- Implemented requirements-based KPI checkers in C++ inside the ADAS Lane Marker Detection application, translating SRD/SRS acceptance thresholds into self-checking pass/fail logic for lane type, color, lateral distance to the rear axle and curvature against ground truth.
- Functionally validated the classical image-processing and DNN (ONNX) lane-detection pipelines on recorded camera and CAN datasets, reusing the same KPI checks to compare both implementations; supported data annotation for the DNN transition.
- Supported ECU bench bring-up with a senior engineer: deployed the C++ binary to target hardware and debugged input-data issues with GDB breakpoints and a JTAG debugger; analyzed vehicle CAN signals in Vector CANoe/CANalyzer.
- Implemented code, branch, and line coverage for perception modules and the KPI framework to expose validation gaps; all development followed MISRA C/C++ compliance.

PROJECTS

UVM Verification of a Set-Associative Write-Back Cache Controller | SystemVerilog, UVM, SVA, Python

- Built a UVM 1.2 environment for a parameterized write-back/write-allocate cache controller with 3 agents (CPU, flush, reactive memory responder via TLM analysis FIFO), a reference-model scoreboard, and SVA protocol checks split into testbench-side and DUT-side assertions.
- Developed 7 directed and constrained-random tests using factory config overrides (backpressure, flush/CPU races, memory error injection); built a Python regression flow merging per-run coverage and reporting holes, with 0 data mismatches across 1,000+ accesses under injected errors.

Functional Verification of LC-3 Processor | UVM/UVMF

- Architected a reusable UVM/UVMF environment covering LC-3 pipeline stages and instruction/data memory; developed sequences, tests, assertions, regression, and coverage models for control-flow, datapath, and corner cases, achieving 100% functional coverage closure.
- Built independent scoreboard-based checking to validate instruction execution against expected pipeline behavior, enabling systematic bug isolation across fetch, decode, and execute stages.

UVM-Based Verification of AXI4 Slave | SystemVerilog

- Built a UVM environment for a 32-bit AXI4 slave supporting FIXED/INCR bursts; implemented agents, scoreboard, SVA protocol checks, and functional coverage for burst types, sizes, and boundary conditions, including constrained-random backpressure scenarios.

Pipelined Transformer Self-Attention Accelerator | SystemVerilog, Design Compiler

- Designed an SRAM-based self-attention accelerator (Q/K/V projections, score and attention matrices); verified against reference results with 217/217 checks passing across 4 test cases and closed timing at 5 ns in Nangate 45 nm.

FIR Filter Accelerator via High-Level Synthesis | SystemC, Catapult HLS, RISC-V Rocket-Chip

- Implemented a 16-tap FIR filter accelerator in SystemC and synthesized it with Catapult HLS at 5 ns, cutting latency from ~3.5K to 69 cycles.

Functional Verification of I2C Multi-Bus Controller | SystemVerilog

- Constructed a layered ncsu_pkg-based verification environment with generator, predictor, scoreboard, assertions, and coverage; exercised protocol behavior and corner cases using functional coverage and SVA, achieving 93% coverage.

EDUCATION

M.S. Computer Engineering | North Carolina State University, Raleigh, NC | CGPA: 3.7/4 **Aug 2024 – May 2026**
Relevant Coursework: Advanced Verification using UVM, ASIC Verification, ASIC and FPGA Design with Verilog, Advanced Microprocessor Architecture, Hardware–Software Codesign

B.Tech. Electronics & Communication Engineering | Dharmsinh Desai University, India **Aug 2018 – May 2022**